

Nursing and Midwifery Council Register linked to Census 2021 – England and Wales: A User Guide

Prof. Iain Atherton and Dr. Michelle Jamieson

Acknowledgements

This project has been enabled by funding from the Economic and Social Research Council (ESRC) and represents collaboration between ADR UK, ADR Scotland, the Nursing and Midwifery Council (NMC) and Office for National Statistics (ONS). We would like to thank Alan Cotterill and Nathan O'Connor (ONS) and Caroline Kenny (NMC) and all their colleagues whose work has made this dataset possible. This work is delivered as part of ADR UK and funded by the ESRC (part of UK Research and Innovation) - ES/W010321.

Introduction

The [Nursing and Midwifery Council](#) (NMC) is the UK's professional regulator for nurses, nursing associates, and midwives. The NMC register holds information on all nurses, midwives and nursing associates that are eligible to practise in the UK collected at initial registration and then at three yearly intervals through revalidation. The register also includes information relating to personal characteristics and qualifications awarded after initial registration. Snapshot data from the NMC register is provided at three timepoints (31 March 2018, 31 March 2021, and 31 March 2022). A further dataset provides indication as to when people left the register between the first two time-points.

A collaboration between the NMC, the Office for National Statistics (ONS) and ADR Scotland has resulted in a linkage of NMC registry data to the 2021 census for England and Wales. The resulting dataset is being made available for approved projects through ONS's [Integrated Data Service](#). Linking the NMC data with the range of social and demographic data provided by the census, enables a range of policy relevant questions to be answered.

This guide provides an overview of the NMC-census linked dataset. It provides information on data provenance and an in-depth overview of the data, upcoming data provisions, and insights into data content, quality, and security measures. It also provides guidance on procedures for data access.

Data sources: An Overview

The NMC Register

The NMC is legally required to hold a register of everyone meeting set standards to practice as a nurse, nursing associate, or midwife in the UK. That makes the register a comprehensive dataset of everyone who is eligible to practise as a nurse, nursing associate or midwife at any one point in time.

People can join the register when they complete an approved course of study or, for those who have qualified overseas, once they can demonstrate they have met a number of requirements, including a test of competence. To remain on the register, [nurses, nursing associates and](#)

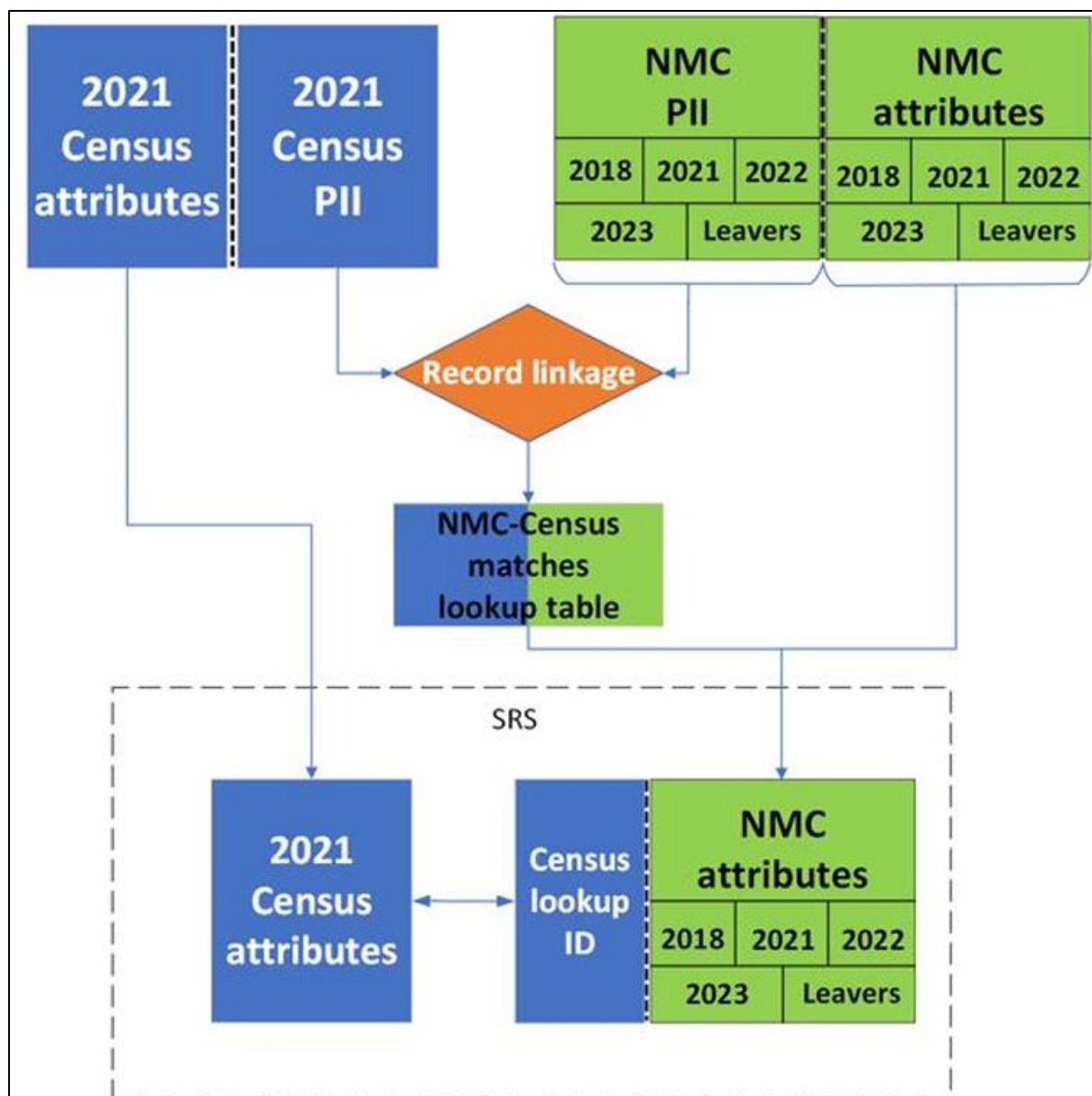
[midwives must revalidate every three years](#). Doing so requires a registrant to provide information demonstrating they have completed at least 450 hours of practice in the preceding three years (900 hours for those with dual nurse and midwife registration) and 35 hours of professional development.

The resulting register is the basis of the Nursing and Midwifery Council Register linked to Census 2021 – England and Wales dataset. More detailed information on variables available is provided in the section on the data dictionary.

The Nursing and Midwifery Council Register linked to Census 2021 – England and Wales dataset overview

ONS have linked extracts from the NMC register to the 2021 census for England and Wales. Figure 1 provides an overview of the structure of the linked dataset. As indicated, three extracts have been used from three timepoints. Everyone on the register as of 31 March of 2018, 2021, 2022 or 2023 are included if they were resident in England and Wales on census night in 2021. A further dataset provides indication as to dates when leavers left the register between March 2018 and 2021. The NMC datasets include everyone on the register at those points in time indicated with the exception of professionals who have chosen to opt out of research in line with its [privacy policy](#).

Figure 1: The Nursing and Midwifery Council Register linked to Census 2021 – England and Wales dataset¹²



¹ Please note, references to PII refer to personally identifiable information

² Diagram courtesy of Nathan O'Connor, Office for National Statistics

The timepoints drawn from the professional register when linked together, as has been done with this data, creates a longitudinal dataset. Thus changes over time can be ascertained including analysis of people entering and leaving the register.

Variables available through the Nursing and Midwifery Council Register linked to Census 2021 – England and Wales dataset

The appendix provides details of variables drawn from the NMC data that are made available in the linked dataset. They include information on a range of topics including demographic and personal characteristics, date of initial registration, country of training, geographic information, employment, and post-registration qualifications.

Any of the census variables made available through ONS's Integrated Data Service can be requested. Information on these variables can be found [here](#). The census provides a snapshot of the entire population of England and Wales every ten years. Information is gathered on issues including demography, housing, health, employment, and unpaid care. Data has been made available by ONS relating to both individuals and for the households in which they reside.

Research potential

Overview

The Nursing and Midwifery Council Register linked to Census 2021 – England and Wales is, to our knowledge, the most comprehensive dataset on the socio-demographics of the nursing and midwifery professions. This data can enable policy relevant questions to be addressed that investigate the social, demographic, and geographic dimensions of nurses, nursing associates and midwives' lives. The three timepoints used from the register, plus 2018-2021

leavers data, enable longitudinal analysis relating to careers. The following provides some indication of what might be done using the register and is in no way exhaustive. Furthermore, comparison of professionals to the wider population would be possible through the census.

Focus on various outcomes are made possible with the Nursing and Midwifery Council Register linked to Census 2021 – England and Wales dataset. Discussions with a stakeholder group, led by ADR Scotland that has supported the development of this resource, has pointed towards specific issues as being of considerable interest, notably recruitment, retention, and professional development. Analysis of the register data can use the NMC variable on date of first registration to develop analysis of those newly on the register. Analysis of those dropping off the register could enable focus on retention. Likely to be closely associated to retention, but a concern in its own right, is the health of registrants. Variables from the census provide indication of [self-assessed general health](#), and [disability](#). Analysis of professional development could utilise data provided by registrants at revalidation on post-registration qualifications.

Research focus

Analysis can focus on the register as a whole, or more likely for the different professional groups represented, nurses, nursing associates or midwives. Focus can be given to the specific nursing branches by disaggregating by adult, children's, mental health, or learning disability nurses, or comparing and contrasting these groups.

There is potential for policy relevant research to look at specific demographic groups. Analysis of those approaching retirement has clear policy relevance given workforce challenges. Men currently make up only a small part of the register with potential to increase numbers, so research that focuses on them and their careers relative to females would be both feasible and of public benefit. Issues relating to diversity is made possible by the data, with variables from both register and census enabling analysis based on age, sex, ethnic identity, and sexual orientation.

Not everyone on the register was necessarily in clinical practice at point of register extracts. The census provides opportunity to ascertain whether people were working at point of census enumeration. If not, indication is provided as to if respondents were looking for work, unable to

work for health reasons and so on. For those who were working, ONS occupational codes give pointers as to the work in which people were engaged in the lead-up to the census. The exemplar project being taken forward is using this data and is briefly outlined in the next subsection.

Examining international migration of professionals is possible with this dataset. Professionals who trained overseas currently make up an increased proportion of the register. Information from the register indicates the country in which professionals trained. The census also has information on country of birth, length of residence, and intention to stay.

Migration within the UK is another possible focus. Moves from one place to another can be analysed through information provided on the register, with changes between entry to the register and revalidations being one means. Census data provides another means with various derived variables being made available (see [here](#)).

The implications of family and caring responsibilities could provide another area of investigation. Derived variables from the census include indication that a person lives in a household where there are [dependent children](#) and the age of that child at point of enumeration. Variables also elucidate where there are [people with a disability in the household](#), and indication of whether that impinges on their activities of daily living to a [greater](#) or [lesser](#) degree. There is also indication as to if the individual [provides unpaid care](#) (including the number of hours done so), or there are [others in the household who do so](#).

Example of research

ADR Scotland researchers who initiated the idea of linking census to registrant data are undertaking an initial project using the Nursing and Midwifery Council Register linked to Census 2021 – England and Wales data to understand the occupations of professionals at point of the 2021 census. This will use occupational data from the census for those on the register in March of the same year. It will disaggregate between nurses, nursing associates, and midwives, and further between branches of nursing. Analysis will investigate what leads to professionals indicating themselves to be in practice and will draw on a range of census and register variables including health, childcare

responsibilities and provision of unpaid care. This focus has been developed through consultation with a stakeholder group that has included representatives from policy, regulator, and practice. They are also using the data to better understand internal migration of registrants within the UK drawing on census data indicating moves in the year prior to enumeration, as well as comparisons of the cross-sections of registrant data.

Data Quality and Utilization

Administrative data

The NMC register was created for the purpose of professional registration rather than for research purposes. These systems are tailored to the administrative needs of maintaining professional standards and registrations within the nursing and midwifery fields. Inevitably this provenance makes this data very different from survey data. As with any administrative dataset not specifically designed for research, analysts need to be aware of potential biases that might arise through approaches to data collection.

Some of the registry variables are only populated at the point of revalidation, which occurs every three years, in particular those providing information on employment and work setting. Even then, data provided about employment and work setting only relates to work professionals have done to meet the 450 practice hours requirement for revalidation. Professionals may well have worked more than these hours and in different roles and settings over the previous years. Or, this may be the sum total of their practice for the preceding period. As it is only provided at the point of revalidation, it could be up to three years' out of date. Further, registrants are not required to provide answers to all questions when registering or at revalidation. Questions on personal characteristics are not mandatory.

As indicated in the appendix, variables relating to where people have worked over the preceding three years might have more than one entry, specifically Employment Type, Scope of Practice, and Work Setting. The length of time in any of those, or the order, is not indicated. Geographical information on the registry is dependent on registrants

typing in their postcode in a free text field. This will not always have been done correctly or as envisaged and so there is scope for error and incomplete data.

Also indicated in the appendix, the NMC age variable is derived from a free text date of birth field entered by registrants. This information may not always have been provided in the correct format or as intended, meaning there is scope for error and incomplete data. Furthermore, the age supplied in the linked dataset is grouped into bands derived from this field and should not be regarded as a precise measure of registrants' chronological age. Issues have also been identified regarding missing records for 2021 and 2023. Whilst proportionately small, the ADR Scotland team are developing approaches to draw on other rounds to enable a more complete dataset.

Professionals are also not required to provide the NMC with information about any nursing or midwifery qualifications awarded after initial registration and what's more, the NMC only accepts information about those [qualifications it recognises](#).

Linkage quality

Record matching has been performed by ONS colleagues in the Data Linkage Hub with the aim of identifying matching data subjects across the NMC register and 2021 census data. There is no common direct identifier in both datasets, so direct record matching is not possible. Instead, personally identifiable information (PII) from the two data sources enabled record linkage using rule-based and score-based matching techniques to identify matching records across the two datasets. PII variables used for record linkage include name, home postcode, work postcode, sex, date of birth.

The ideal would be to match every NMC record to a corresponding census record, however without an existing common identifier, limitations arise which affect the linkage quality. These limitations include, but are not limited to: missingness in PII variables in either dataset (e.g. some questions in the NMC register are optional); poor quality PII data or recording/formatting inconsistencies between the datasets. As well as quality errors and inconsistencies, the NMC register and census do not necessarily have the same coverage of records (e.g. NMC registrants may not have been living in

England or Wales at the time of the census). This is another reason why not all registrants will have a linked census record.

The linkage provided links to the census for 83.82% of NMC register records. The precision (accuracy) – the proportion of links that were estimated to be true matches – is 99.95%. The recall (sensitivity) – the proportion of true matches that were identified in the linked data – was estimated at 94.87%.

The PII used for linkage is not accessible outside controlled project areas in ONS' secure Data Access Platform. None of this identifiable data will be made available to researchers in the SRS or otherwise. Disclosure control measures are taken at multiple steps in the data journey to ensure that none of the data subjects can be personally identified by users of the linked dataset.

Data Access

Overview

This dataset is held and accessed through SRS. This is a facility that is accredited under the Digital Economy Act (DEA) 2017. This enables the data to be held and accessed securely for approved projects and by accredited researchers. It follows the principles of the [Five Safes](#) to ensure that data is used in a safe and secure manner: safe people; safe projects; safe settings; safe outputs; and safe data. These are enabled by policies, infrastructure and training that will be central to any project using this or other ADR UK Flagship Data. Enabling research with secure data is at the heart of the SRS as set out [here](#).

Becoming an accredited researcher

Only accredited users with accredited projects can access the data. This involves the completion of a half-day course and passing an associated test. More detailed information including where safe training is offered can be found [here](#). This accreditation has to be updated every five years through retaking the course and test.

Applications to use data

Applying to use the data involves the submission of a proposal to the SRS. Details of the process involved are provided [here](#). It involves an initial expression of interest which enables the engagement team to get in touch and discuss what you are envisaging with you. Assuming the proposal is feasible, then a more substantial application will set out the project, including its aims, methods, who will be involved, dissemination plans, and – importantly – the public benefit. The proposal is then reviewed by a Research Accreditation Panel and by the NMC.

Accessing the data

An Assured Organisational Connectivity Agreement (AOC) enables a researcher to access data from their university or remote office space. An AOC is an agreement that gives assurances that an organisation has necessary controls and processes in place and takes responsibility for that access. Organisations without this agreement in place can apply. More details on access and AOCs can be found [here](#).

Software

Details of tools available to researchers using the SRS can be found [here](#). As of September 2025, software available includes: Adobe Acrobat Reader DC, Python 3.12, QGIS 3.28.11, Jupyter Notebook, JupyterLab, Microsoft Office Professional Plus 2021, ML-Win (3.04), Notepad++, Qtconsole, R for Windows (4.4), R Studio (2023.06.1), SPSS v29, Spyder 6, STATA (16MP, 17SE, 18MP), Winzip, MPlus 8 Demo, SQL Server Management Studio, Java Development Kit.

Feedback

Users are encouraged to provide feedback on documentation through emailing Iain Atherton (i.atherton@napier.ac.uk). Reflections on areas where further information would be helpful or clarity would be valued.

Nursing & Midwifery Population Data Research Community

A community of interest is in development that aims to bring together those whose work focuses on the nursing and midwifery professions and specifically using nationally representative data. Analysts whether from universities, government or otherwise are encouraged to join, as are those whose interest is in using the data. The NMC-Census Dataset 2021 will be a major focus though it will also support work using complimentary datasets. Its intention is to support and encourage the use of administrative data to the benefit of nursing and midwifery professions (and hence the wider public), and to inform the development of new resources.

More information can be obtained through emailing Iain Atherton (i.atherton@napier.ac.uk).

Further information

For queries about the Nursing and Midwifery Council Register linked to Census 2021 – England and Wales dataset, please contact SRS customer support using the following email: srs.customer.support@ons.gov.uk.

Appendix – Register Data Dictionary

Variable Name	Variable Definition	Data Type	Values	Mandatory/Optional data population	Additional Notes
nmc_census_id	Unique identifier	String		Mandatory	This identifier enables the NMC tables to be joined to Census. For any records that are not linked to Census, this column will be populated with a unique hashed ID
Registration Date	Date of first registration to the NMC	Date		Mandatory	
Registration Type	Type of registration	String	Midwife (incl SCPHN); Nurse (incl.SCPHN); Nurse and Midwife (incl SCPHN); Nursing Associate	Mandatory	
FieldOfNursing	The field/s of nursing qualified in	String	Adult nurse; Mental health nurse; Learning disabilities nurse; Children's nurse	Mandatory	People can be registered in more than one field of nursing for example both adult and mental health nursing

Residenceoa01	Residence as ONS 2001 Census Output Areas	String	E00, W00, S00, 95AA010001-95ZZ519999, pseudo or null		
Residenceoa11	Residence as ONS 2011 Census Output Areas	String	E00, W00, S00, N00, pseudo or null		
Residenceoa21	Residence as ONS 2021 Census Output Areas	String	E00, W00, N20, pseudo or null		
ResidenceCountry	Country of residence	String		Optional	
ResidenceCounty	UK county of residence	String		Optional	
TrainingCountry	Country of initial training for the professional	String		Optional	
AgeGroup	Professional's age in 5 year age groups	String	Under 30 years; 30-39 years; 40-49 years; 50-54 years; 55-59 years; 60-64 years; 65-69 years; 70-74 years; 75 years and over	Mandatory	
Gender	Professional's gender	String	Female; Male; Unknown	Optional	
Disability	Professional's disability	String	Yes; No; Prefer not to say; Unknown	Optional	
ReligionandBelief	Professional's religion or belief	String	Buddhist; Christian; Hindu; Jewish; Muslim; Sikh; Other; None; Prefer not to say; Unknown	Optional	

SexualOrientation	Professional's sexual orientation	String	Bisexual; Gay or lesbian; heterosexual or straight; Other; Prefer not to say; Unknown	Optional	
Ethnicity	Professional's ethnicity	String	Arab; Asian-Any other Asian background; Asian-Bangladeshi; Asian-Chinese; Asian-Filipina/Filipino; Asian-Indian; Asian-Pakistani; Black-African; Black-Any other Black background; Black-Caribbean; Mixed-Any other Mixed background; Mixed-White and Asian; Mixed-White and Black African; Mixed-White and Black Caribbean; Other-Any other ethnic group; Prefer not to say; Unknown; White-Any other White background; White-English/Welsh/Scottish/Northern Irish/British; White-Gypsy or Irish Traveller; White-Irish	Optional	
GenderIdentity	Professional's gender identity - does gender identity	String	Yes; No; Prefer not to say; Unknown	Optional	

	match sex as registered at birth				
EmploymentType	Employment type/s declared at most recent revalidation	String	Employed directly; employed via agency; self-employed; volunteering	Mandatory	This information is only captured at the point of revalidation, which is every three years. It is therefore available only for the professionals who have been through revalidation. People are required to declare the jobs they have done over the previous three years to meet the 450 practice hours requirement of revalidation. People can include details for multiple employment. We have included information provided for the individual's most recent revalidation up to the date of data extraction (31 March 2018). The most recent revalidation date will differ for each person as it depends on the length of time that they have been in the

					NMC register and also when they joined. Revalidation was introduced in 2016.
Employeroa01	Place of employment as ONS 2001 Census Output Areas	String	E00, W00, S00, 95AA010001-95ZZ519999, pseudo or null	Mandatory	This information is only captured at the point of revalidation, which is every three years. It is therefore available only for the professionals who have been through revalidation. People are required to declare the jobs they have done over the previous three years to meet the 450 practice hours requirement of revalidation. People can include details for multiple employment. We have included information provided for the individual's most recent revalidation up to the date of data extraction (31 March 2018). The most

					recent revalidation date will differ for each person at it depends on the length of time that they have been in the NMC register and also when they joined. Revalidation was introduced in 2016.
Employeroa11	Place of employment as ONS 2011 Census Output Areas	String	E00, W00, S00, N00, pseudo or null		
Employeroa21	Place of employment as ONS 2021 Census Output Areas	String	E00, W00, N20, pseudo or null		
ScopeOfPractice	Scope/s of practice declared at most recent revalidation	String	Commissioning; Direct clinical care or management – adult and general care nursing; Direct clinical care or management – children’s and neo-natal nursing; Direct clinical care or management – health visiting; Direct clinical care or management – learning disabilities nursing; Direct clinical care or management –	Mandatory	This information is only captured at the point of revalidation, which is every three years. It is therefore available only for the professionals who have been through revalidation. People are required to declare the jobs they have done over the previous three years to meet the 450 practice hours requirement of

			mental health nursing; Direct clinical care or management – midwifery; Direct clinical care or management – occupational health; Direct clinical care or+F21 management – public health: Direct clinical care or management – school nursing; Direct clinical care or management – other; Education; Policy; Quality assurance or inspection; Research; Other		revalidation. People can include details for multiple employment. We have included information provided for the individual's most recent revalidation up to the date of data extraction (31 March 2018). The most recent revalidation date will differ for each person as it depends on the length of time that they have been in the NMC register and also when they joined. Revalidation was introduced in 2016.
WorkSetting	Work setting/s declared at most recent revalidation	String	Ambulance service; Care home sector; Community setting, including district nursing and community psychiatric nursing; Consultancy; Cosmetic or aesthetic sector; Governing body or other leadership; GP practice or other primary care; Hospital or other secondary care; Inspectorate or regulator;	Optional	This information is only captured at the point of revalidation, which is every three years. It is therefore available only for the professionals who have been through revalidation. People are required to declare the jobs they have done over the previous three years to meet the 450 practice

			Insurance or legal; Maternity unit or birth centre; Military; Occupational health; Police; Policy organisation; Prison; Private domestic setting; Public health organisation; School; Specialist or other tertiary care including hospice; Telephone or e-health advice; Trade union or professional body; University or other research facility; Voluntary or charity sector; Other		hours requirement of revalidation. People can include details for multiple employment. We have included information provided for the individual's most recent revalidation up to the date of data extraction (31 March 2018). The most recent revalidation date will differ for each person as it depends on the length of time that they have been in the NMC register and also when they joined. Revalidation was introduced in 2016.
--	--	--	---	--	--

PostRegQualifications	Qualification registered post initial registration	String	Various codes indicate qualifications that are no longer relevant. That remain relevant are the following: PAMD PBNN PCCH PCCN PCCP PCCS PCND PCOP PCOR PCPP PCPS PD01 PD02 PD03 PD04 PD05 PD06 PD09 PD11 PDAM PDAS PDDN PDES PDHN PDHS PDIC PDIM PDIN PDMW PDNI PDNS PDNW PDOP PDOR PDPI PDPN PDPP PDPS PDSA PDS PGPN	Optional	Once registered, professionals can declare additional qualifications they have done with the NMC. The NMC only recognise certain qualifications however.
-----------------------	--	--------	---	----------	--

			PHNO PHVA PHVD PHVF PICN PNAI PNBS PNCB PNCN PNMH PNNN POHN PPBS PPCB PPCC PPCN PPPS PSCN RFHN: Family health nurse RHSV: Health visitor RM: Midwife RN1: Adult nurse, level 1 RN2: Adult nurse, level 2 RN3: Mental health nurse, level 1 RN4: Mental health nurse, level 2 RN5: Learning disabilities nurse, level 1 RN6: Learning disabilities nurse, level 2 RN7: General nurse, level 2 RN8: Children's nurse, level 1 RN9: Fever nurse, level 2 RNA: Adult nurse, level 1 RNC: Children's nurse, level 1 RNLD: Learning disabilities		
--	--	--	--	--	--

			nurse, level 1 RNMH: Mental health nurse, level 1 ROH: Occupational health nurse RPHN: Specialist community public health nurse RSN: School nurse S001 S004 S005 S006 S009 S010 S011 S012 S013 S014 S016 S017 S018 S025 S029 S030 S032 SCLD: Specialist practitioner: Community learning disabilities nursing SCMH: Specialist practitioner: Community mental health nursing SN: School nurse SPA: Specialist practitioner: Adult nursing SPC: Specialist practitioner: Children's nursing		
--	--	--	--	--	--

			SPCB SPCC: Specialist practitioner: Community children's nursing SPCH SPCLD SPCMH SPDN: Specialist practitioner: District nursing SPEC SPGI SPGP: Specialist practitioner: General practice nursing SPHV SPIC SPLD: Specialist practitioner: Learning disability nurse SPMH: Specialist practitioner: Mental health SPOH SPON SPPC SPPH SPQG SPSN T100 T105 T115 T120 T125 T130 T135 T140 T145 T150 T155 T160 T165 T170 T175 T180 T185 T190		
--	--	--	--	--	--

			T195 T200 T330 T345 T515 T530 T575 T710 T715 T730 T740 T745 T775 T780 T790 T795 T800 T920 T930 T940 T945 T975 T990 TAAH TABD TACM TADE TADM TBAB TBAD TBAH TBAM TBAN TBBM TBSC TCAN TCBC TCCC TCCM TCCN TCCO TCCP TCCR TCCS TCDC TCDN TCEC TCFC TCGC TCGP TCH: Teacher		
--	--	--	--	--	--

			TCHC TCHO TCHV TCMH TCMN TCOH TCPN TCSN TDAC TDCN TDES TFAD TFAH TFAM TFAN TFBC TFEN TGAD TGAH TGAM TGAN TGBN THAD THAH THAM THAN THBD THBM THBN THCD THCN THDN THEN THVA TJAN TKAN TLAN TMAM TMAN TMCN TMHN TMTA TNAV TNTA TNTB TOHN TPPN TXXN TYAN TYBC TYBM		
--	--	--	--	--	--

			TYBN TYDN TYED TYEN TYHH TYHN TYOD TYOH TYOM TYON V100: Community practitioner nurse prescriber V100-EXP V150: Community practitioner nurse prescriber (without SPQ or SCPHN) V200: Nurse independent prescriber (extended formulary) V300: Nurse independent / supplementary prescriber		
--	--	--	---	--	--